



M003 Week 5 — English Worksheet (Advanced)

Topic: Recreate the Statues (x, y, z) · **Course:** 3D Coordinates · **Level:** Advanced · **Time:** about 45 minutes

A statue is many **stacked fill boxes**: two legs, a torso, a head, and two arms. One **fill** makes a solid box between **two corners**, each written **(x, y, z)** — x **across**, y **up**, z **forward**. To make a real figure you have to plan: where each part **starts** (its bottom y), how **wide** it is (x and z), and that the feet sit on the **ground (y = 0)**. Read carefully — a part can hide inside another, or float above the floor.

● Stand on your **home spot** (red block) every run. Move your feet, move your head.

1 · Predict 🤖

Read each box. Write what you will see, **where it sits** on the statue, **and the reason**. Do **not** run it.

```
fill GRAY from (3, 0, 4) to (3, 2, 4)
fill GRAY from (5, 0, 4) to (5, 2, 4)
```

Two thin boxes, x = 3 and x = 5, both y = 0–2. What part is this, why two of them, and why y = 0?


```
fill GRAY from (2, 3, 4) to (2, 5, 4)
fill GRAY from (6, 3, 4) to (6, 5, 4)
```

These start at y = 3 and sit at x = 2 and x = 6 — outside the legs. What part is this, and why is the x outside?


```
fill WHITE from (3, 7, 4) to (5, 8, 4)
```

This is the highest box ($y = 7-8$). What part is it, and how do you know from the y?


```
fill GRAY from (3, 3, 4) to (5, 6, 4)  
fill WHITE from (3, 7, 4) to (5, 8, 4)
```

Two boxes together. The torso ends at $y = 6$ but the head starts at $y = 7$. What sits between them, and what does that gap mean?

2 · Spot the Bug 🐛

Each block is broken. Read what it should do, write the fixed code, then explain.

Bug A — The torso should sit **on top** of the legs. The legs go $y = 0-2$, so the torso should start at $y = 3$. Right now the torso starts at $y = 0$, so it builds over the legs and nothing rises.

```
fill GRAY from (3, 0, 4) to (5, 6, 4)
```

Hint: the **2nd number (y)** is height. Where should the torso's bottom y be?

Write the fixed code:

Why was it wrong? Why does your fix work? (2–3 sentences.)

Bug B — This should be the **right arm**, hanging on the **side** of the torso at $x = 6$. The torso fills $x = 3-5$, so $x = 4$ is **inside** it — the arm disappears into the body.

```
fill GRAY from (4, 3, 4) to (4, 5, 4)
```

Hint: to be on the outside, the arm's **1st number (x)** must be past the torso's $x = 3-5$.

Write the fixed code:

Why was it wrong? Why does your fix work? (2–3 sentences.)

Bug C (*the tricky one*) — Your friend says their whole statue is **floating** — there is a gap of air under the feet. The legs are built at **y = 1–3** instead of touching the ground. The bottom of the feet must be on the floor.

```
fill GRAY from (3, 1, 4) to (3, 3, 4)
fill GRAY from (5, 1, 4) to (5, 3, 4)
```

Hint: the ground is **y = 0**. What is the lowest y a foot should have?

Write the fixed code:

Why was it wrong? Why does your fix work? (2–3 sentences.)

3 · Build It 📷

Walk around a statue in your world. Then plan and build a **full golem** on your home spot — **6 fill boxes**: two legs, a torso, two arms, and a head.

📊 Recipe (every corner is **(x, y, z)**, feet on the ground $y = 0$): - left leg: gray from (3, 0, 4) to (3, 2, 4) - right leg: gray from (5, 0, 4) to (5, 2, 4) - torso: gray from (3, 3, 4) to (5, 6, 4) - left arm: gray from (2, 3, 4) to (2, 5, 4) - right arm: gray from (6, 3, 4) to (6, 5, 4) - head: white from (3, 7, 4) to (5, 8, 4)

Before you build, **plan the heights** in the space below: which y does each part start and stop at?

🌟 **Modify challenge:** after it works, change one thing of your own — a new colour, a different pose, or an extra part (a hat, a tail, longer arms). Write what you changed.

Then build, and send a **photo or video**. Finish these — 5 to 6 sentences.

I planned the heights like this: legs $y = \dots$, torso $y = \dots$, head $y = \dots$

The feet touch the ground because the lowest **y** is ...

My arms sit on the outside because their **x** is ...

Each part stacks on the one below because its **y** starts at ...

One part I had to fix was ... because ...

The thing I changed for the modify challenge was ...

4 · Explain It 🎥

Film yourself showing your statue. Try to use these words: **fill, box, x, y, z, stack, ground, home spot**.

1. Show your home spot and stand on it. Say why you stand there every run.
2. Walk around your statue — show the front, a side, and the back.
3. Point at one part and read its corners: "from x ..., y ..., z ... to x ..., y ..., z ...".
4. Say in your own words how you know the head ends up on top and the feet stay on the ground.

Write a short paragraph (5–6 sentences) explaining how your statue is built. Then answer: if you taught a friend, what would you teach first, and why?

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.